Using this arrow representation, sketch each of the following vector fields:

(a) $\mathbf{E}_1 = -\hat{\mathbf{x}}y$

(b) $\mathbf{E}_2 = \hat{\mathbf{y}}x$

(c) $\mathbf{E}_3 = \hat{\mathbf{x}}x + \hat{\mathbf{y}}y$

(d) $\mathbf{E}_4 = \hat{\mathbf{x}}x + \hat{\mathbf{y}}2y$

(e) $\mathbf{E}_5 = \hat{\boldsymbol{\phi}}r$

(f) $\mathbf{E}_6 = \hat{\mathbf{r}}\sin\phi$

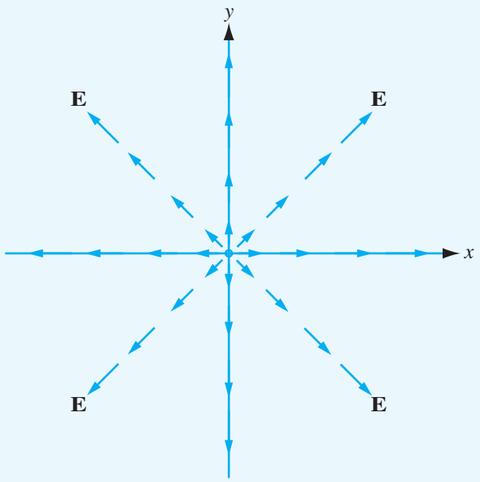


Figure P3.20 Arrow representation for vector field $\mathbf{E} = \hat{\mathbf{r}}\,r$ (Problem 3.20).

3.21 Use arrows to sketch each of the following vector fields:

(a) $\mathbf{E}_1 = \hat{\mathbf{x}}x - \hat{\mathbf{y}}y$

(b) $\mathbf{E}_2 = -\hat{\boldsymbol{\phi}}$

(c) $\mathbf{E}_3 = \hat{\mathbf{y}}\,(1/x)$

(d) $\mathbf{E}_4 = \hat{\mathbf{r}}\cos\phi$

Sections 3-2 and 3-3: Coordinate Systems

3.22 Convert the coordinates of the following points from Cartesian to cylindrical and spherical coordinates:

*(a) $P_1 = (1,2,0)$

(b) $P_2 = (0,0,2)$

(c) $P_3 = (1,1,3)$

*(d) $P_4 = (-2,2,-2)$

3.23 Convert the coordinates of the following points from cylindrical to Cartesian coordinates:

(a) $P_1 = (2,\pi/4,-3)$

(b) $P_2 = (3,0,-2)$

(c) $P_3 = (4,\pi,5)$

3.24 Convert the coordinates of the following points from spherical to cylindrical coordinates:

*(a) $P_1 = (5,0,0)$

(b) $P_2 = (5,0,\pi)$

(c) $P_3 = (3,\pi/2,0)$

3.25 Use the appropriate expression for the differential surface area $d\mathbf{s}$ to determine the area of each of the following surfaces:

(a) $r = 3;\ 0 \le \phi \le \pi/3;\ -2 \le z \le 2$

(b) $2 \le r \le 5;\ \pi/2 \le \phi \le \pi;\ z = 0$

*(c) $2 \le r \le 5;\ \phi = \pi/4;\ -2 \le z \le 2$

(d) $R = 2;\ 0 \le \theta \le \pi/3;\ 0 \le \phi \le \pi$

(e) $0 \le R \le 5;\ \theta = \pi/3;\ 0 \le \phi \le 2\pi$

Also sketch the outline of each surface.

3.26 Find the volumes described by the following:

*(a) $2 \le r \le 5;\ \pi/2 \le \phi \le \pi;\ 0 \le z \le 2$

(b) $0 \le R \le 5;\ 0 \le \theta \le \pi/3;\ 0 \le \phi \le 2\pi$

Also sketch the outline of each volume.

3.27 A section of a sphere is described by $0 \le R \le 2$, $0 \le \theta \le 90°$, and $30° \le \phi \le 90°$. Find the following:

(a) The surface area of the spherical section.

(b) The enclosed volume.

Also sketch the outline of the section.

3.28 A vector field is given in cylindrical coordinates by

$$\mathbf{E} = \hat{\mathbf{r}}r\cos\phi + \hat{\boldsymbol{\phi}}r\sin\phi + \hat{\mathbf{z}}z^2.$$

Point $P = (2,\pi,3)$ is located on the surface of the cylinder described by $r = 2$. At point P, find:

(a) The vector component of $\mathbf{E}$ perpendicular to the cylinder.

(b) The vector component of $\mathbf{E}$ tangential to the cylinder.

3.29 At a given point in space, vectors $\mathbf{A}$ and $\mathbf{B}$ are given in spherical coordinates by

$$\mathbf{A} = \hat{\mathbf{R}}4 + \hat{\boldsymbol{\theta}}2 - \hat{\boldsymbol{\phi}},$$

$$\mathbf{B} = -\hat{\mathbf{R}}2 + \hat{\boldsymbol{\phi}}3.$$

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使用此箭头表示，绘制以下每个向量场的草图：

(a) $E_1 = -\hat{x}y$ (b)
$E_2 = \hat{y}x$ (c) E
$_3 = \hat{x}x + \hat{y}y$ (d) E
$_4 = \hat{x}x + \hat{y}2y$ (e)
$E_5 = \hat{\boldsymbol{\phi}}r$ (f) $E_6 = \hat{r}$
$\sin\phi$

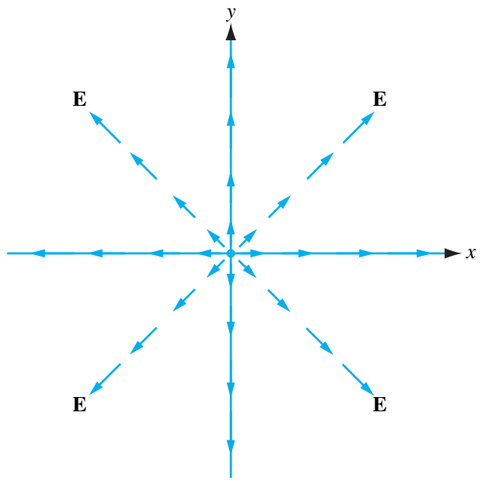


图 P3.20 矢量场 E = r̂r 的箭头表示（问题 3.20）。

3.21 用箭头画出以下每个向量场：

(a) $E_1 = \hat{x}x - \hat{y}y$
(b) $E_2 = -\hat{\boldsymbol{\phi}}$c) E
$(_3 = \hat{y}\,(1/x)$ (d) E
$_4 = \hat{r}\cos\phi$

第 3-2 节和第 3-3 节：坐标系

3.22 将下列点的坐标从笛卡尔坐标转换为柱坐标和球坐标：

*(a) $P_1 = (1, 2, 0)$b) $P_2 = (0, 0, 2)$c) $P_3 = (1, 1, 3)$
((

*(d) $P_4 = (-2,2,-2)$

3.23 将下列点的坐标从圆柱坐标转换为笛卡尔坐标：

(a) $P_1 = (2、\pi/4、-3)$b) $P_2 = (3、0、-2)$c) $P_3 = (4、\pi、5)$
((

3.24 将下列点的坐标从球坐标转换为柱坐标：

*(a) $P_1 = (5, 0, 0)$b) $P_2 = (5, 0, \pi)$c) $P_3 = (3, \pi/2, 0)$
((

3.25 使用微分表面积 ds 的适当表达式来确定以下每个表面的面积：

(a) $r = 3;\ 0 \le \phi \le \pi/3;\ -2 \le z \le 2$ (
b) $2 \le r \le 5;\ \pi/2 \le \phi \le \pi;\ z = 0$

*(c) $2 \le r \le 5;\ \phi = \pi/4;\ -2 \le z \le 2$ (
d) $R = 2;\ 0 \le \theta \le \pi/3;\ 0 \le \phi \le \pi$e) 0
($\le R \le 5;\ \theta = \pi/3;\ 0 \le \phi \le 2\pi$

还勾勒出每个表面的轮廓。

3.26 找到下列描述的卷：

*(a) $2 \le r \le 5;\ \pi/2 \le \phi \le \pi;\ \ 0 \le z \le 2$ (
b) $0 \le R \le 5;\ 0 \le \theta \le \pi/3;\ 0 \le \phi \le 2\pi$

还勾勒出每卷的轮廓。

3.27 球体的截面由 $0 \le R \le 2$、$0 \le \theta \le 90°$ 和 $30° \le \phi \le 90°$ 描述。找到以下内容：

(a) 球形截面的表面积。

(b) 封闭体积。

还勾勒出该部分的轮廓。

3.28 向量场在柱坐标系中由下式给出

$$\mathbf{E} = \hat{\mathbf{r}}r\cos\phi + \hat{\boldsymbol{\phi}}r\sin\phi + \hat{\mathbf{z}}z^2.$$

点 $P = (2、\pi、3)$ 位于 $r = 2$ 描述的圆柱体表面上。在点 P 处，找到：

(a) E 垂直于圆柱体的矢量分量。

(b) E 与圆柱体相切的矢量分量。

3.29 在空间中的给定点，向量 A 和 B 在球坐标中给出：

$$\mathbf{A} = \hat{\mathbf{R}}4 + \hat{\boldsymbol{\theta}}2 - \hat{\boldsymbol{\phi}},$$

$$\mathbf{B} = -\hat{\mathbf{R}}2 + \hat{\boldsymbol{\phi}}3.$$

Find:

(a) The scalar component, or projection, of **B** in the direction of **A**.

(b) The vector component of **B** in the direction of **A**.

(c) The vector component of **B** perpendicular to **A**.

*3.30 Given vectors

$$\mathbf{A} = \hat{\mathbf{r}}(\cos\phi + 3z) - \hat{\boldsymbol{\phi}}(2r + 4\sin\phi) + \hat{\mathbf{z}}(r - 2z),$$
$$\mathbf{B} = -\hat{\mathbf{r}}\sin\phi + \hat{\mathbf{z}}\cos\phi,$$

find

(a) θ_{AB} at $(2, \pi/2, 0)$

(b) A unit vector perpendicular to both **A** and **B** at $(2, \pi/3, 1)$

3.31 Find the distance between the following pairs of points:

(a) $P_1 = (1,2,3)$ and $P_2 = (-2,-3,-2)$ in Cartesian coordinates.

(b) $P_3 = (1, \pi/4, 3)$ and $P_4 = (3, \pi/4, 4)$ in cylindrical coordinates.

(c) $P_5 = (4, \pi/2, 0)$ and $P_6 = (3, \pi, 0)$ in spherical coordinates.

3.32 Determine the distance between the following pairs of points:

*(a) $P_1 = (1,1,2)$ and $P_2 = (0,2,3)$

(b) $P_3 = (2, \pi/3, 1)$ and $P_4 = (4, \pi/2, 3)$

(c) $P_5 = (3, \pi, \pi/2)$ and $P_6 = (4, \pi/2, \pi)$

3.33 Transform the vector

$$\mathbf{A} = \hat{\mathbf{R}}\sin^2\theta\cos\phi + \hat{\boldsymbol{\theta}}\cos^2\phi - \hat{\boldsymbol{\phi}}\sin\phi$$

into cylindrical coordinates and then evaluate it at $P = (2, \pi/2, \pi/2)$.

3.34 Transform the following vectors into cylindrical coordinates and then evaluate them at the indicated points:

(a) $\mathbf{A} = \hat{\mathbf{x}}(x+y)$ at $P_1 = (1,2,3)$

(b) $\mathbf{B} = \hat{\mathbf{x}}(y-x) + \hat{\mathbf{y}}(x-y)$ at $P_2 = (1,0,2)$

*(c) $\mathbf{C} = \hat{\mathbf{x}}y^2/(x^2+y^2) - \hat{\mathbf{y}}x^2/(x^2+y^2) + \hat{\mathbf{z}}4$ at $P_3 = (1,-1,2)$

(d) $\mathbf{D} = \hat{\mathbf{R}}\sin\theta + \hat{\boldsymbol{\theta}}\cos\theta + \hat{\boldsymbol{\phi}}\cos^2\phi$ at $P_4 = (2, \pi/2, \pi/4)$

*(e) $\mathbf{E} = \hat{\mathbf{R}}\cos\phi + \hat{\boldsymbol{\theta}}\sin\phi + \hat{\boldsymbol{\phi}}\sin^2\theta$ at $P_5 = (3, \pi/2, \pi)$

3.35 Transform the following vectors into spherical coordinates and then evaluate them at the indicated points:

(a) $\mathbf{A} = \hat{\mathbf{x}}y^2 + \hat{\mathbf{y}}xz + \hat{\mathbf{z}}4$ at $P_1 = (1,-1,2)$

(b) $\mathbf{B} = \hat{\mathbf{y}}(x^2+y^2+z^2) - \hat{\mathbf{z}}(x^2+y^2)$ at $P_2 = (-1,0,2)$

*(c) $\mathbf{C} = \hat{\mathbf{r}}\cos\phi - \hat{\boldsymbol{\phi}}\sin\phi + \hat{\mathbf{z}}\cos\phi\sin\phi$ at $P_3 = (2, \pi/4, 2)$

(d) $\mathbf{D} = \hat{\mathbf{x}}y^2/(x^2+y^2) - \hat{\mathbf{y}}x^2/(x^2+y^2) + \hat{\mathbf{z}}4$ at $P_4 = (1,-1,2)$

Sections 3-4 to 3-7: Gradient, Divergence, and Curl Operators

3.36 Find the gradient of the following scalar functions:

(a) $T = 3/(x^2+z^2)$

(b) $V = xy^2z^4$

(c) $U = z\cos\phi/(1+r^2)$

(d) $W = e^{-R}\sin\theta$

*(e) $S = 4x^2e^{-z} + y^3$

(f) $N = r^2\cos^2\phi$

(g) $M = R\cos\theta\sin\phi$

3.37 For each of the following scalar fields, obtain an analytical solution for ∇T and generate a corresponding arrow representation.

(a) $T = 10 + x$, for $-10 \le x \le 10$

*(b) $T = x^2$, for $-10 \le x \le 10$

(b) $T = 100 + xy$, for $-10 \le x \le 10$

(c) $T = x^2y^2$, for $-10 \le x, y \le 10$

(e) $T = 20 + x + y$, for $-10 \le x, y \le 10$

(f) $T = 1 + \sin(\pi x/3)$, for $-10 \le x \le 10$

*(g) $T = 1 + \cos(\pi x/3)$, for $-10 \le x \le 10$

(h) $T = 15 + r\cos\phi$, for $\begin{cases} 0 \le r \le 10 \\ 0 \le \phi \le 2\pi. \end{cases}$

(i) $T = 15 + r\cos^2\phi$, for $\begin{cases} 0 \le r \le 10 \\ 0 \le \phi \le 2\pi. \end{cases}$

*3.38 The gradient of a scalar function T is given by

$$\nabla T = \hat{\mathbf{z}}e^{-3z}$$

If $T = 10$ at $z = 0$, find $T(z)$.

3.39 Follow a procedure similar to that leading to Eq. (3.82) to derive the expression given by Eq. (3.83) for ∇ in spherical coordinates.

*3.40 For the scalar function $V = xy^2 - z^2$, determine its directional derivative along the direction of vector $\mathbf{A} = (\hat{\mathbf{x}} - \hat{\mathbf{y}}z)$ and then evaluate it at $P = (1,-1,4)$.

3.41 Evaluate the line integral of $\mathbf{E} = \hat{\mathbf{x}}x - \hat{\mathbf{y}}y$ along the segment P_1 to P_2 of the circular path shown in Fig. P3.41.

寻找：

(a) B 在 A 方向上的标量分量或投影。

(b) B 在 A 方向上的矢量分量。

(c) B 垂直于 A 的矢量分量。

*3.30 给定向量

$$\mathbf{A} = \hat{\mathbf{r}}(\cos\phi + 3z) - \hat{\boldsymbol{\phi}}(2r + 4\sin\phi) + \hat{\mathbf{z}}(r - 2z),$$
$$\mathbf{B} = -\hat{\mathbf{r}}\sin\phi + \hat{\mathbf{z}}\cos\phi,$$

找到

(a) $(2, \pi/2, 0)$ 处的 θ_{AB} (b) $(2, \pi/3, 1)$ 处垂直于 A 和 B 的单位向量 3.31 求下列两点之间的距离：

(a) 笛卡尔坐标中的 $P_1 = (1, 2, 3)$ 和 $P_2 = (-2, -3, -2)$。

(b) 柱坐标中的 $P_3 = (1、\ \pi/4、\ 3)$ 和 $P_4 = (3、\ \pi/4、\ 4)$。

(c) 球坐标中的 $P_5 = (4、\ \pi/2、\ 0)$ 和 $P_6 = (3、\ \pi、\ 0)$。

3.32 确定下列两点之间的距离：

*(a) $P_1 = (1, 1, 2)$ 和 $P_2 = (0, 2, 3)$

(b) $P_3 = (2、\ \pi/3、\ 1)$ 和 $P_4 = (4、\ \pi/2、\ 3)$

(c) $P_5 = (3、\ \pi、\ \pi/2)$ 和 $P_6 = (4、\ \pi/2、\ \pi)$

3.33 变换向量

$$\mathbf{A} = \hat{\mathbf{R}}\sin^2\theta\cos\phi + \hat{\boldsymbol{\theta}}\cos^2\phi - \hat{\boldsymbol{\phi}}\sin\phi$$

转换为柱坐标，然后在 $P = (2、\ \pi/2、\ \pi/2)$ 处对其进行评估。

3.34 将下列向量变换为柱坐标，然后在指定点处对它们求值：

(a) $\mathbf{A} = \hat{\mathbf{x}}(x+y)$ at $P_1 = (1,2,3)$

(b) $\mathbf{B} = \hat{\mathbf{x}}(y-x) + \hat{\mathbf{y}}(x-y)$ at $P_2 = (1,0,2)$

*(c) $\mathbf{C} = \hat{\mathbf{x}}y^2/(x^2+y^2) - \hat{\mathbf{y}}x^2/(x^2+y^2) + \hat{\mathbf{z}}4$ at $P_3 = (1,-1,2)$

(d) $\mathbf{D} = \hat{\mathbf{R}}\sin\theta + \hat{\boldsymbol{\theta}}\cos\theta + \hat{\boldsymbol{\phi}}\cos^2\phi$ at $P_4 = (2, \pi/2, \pi/4)$

*(e) $\mathbf{E} = \hat{\mathbf{R}}\cos\phi + \hat{\boldsymbol{\theta}}\sin\phi + \hat{\boldsymbol{\phi}}\sin^2\theta$ at $P_5 = (3, \pi/2, \pi)$

3.35 将下列向量变换为球坐标，然后在指定点对其求值：

(a) $\mathbf{A} = \hat{\mathbf{x}}y^2 + \hat{\mathbf{y}}xz + \hat{\mathbf{z}}4$ at $P_1 = (1,-1,2)$

(b) $\mathbf{B} = \hat{\mathbf{y}}(x^2+y^2+z^2) - \hat{\mathbf{z}}(x^2+y^2)$ at $P_2 = (-1,0,2)$

*(c) $\mathbf{C} = \hat{\mathbf{r}}\cos\phi - \hat{\boldsymbol{\phi}}\sin\phi + \hat{\mathbf{z}}\cos\phi\sin\phi$ at $P_3 = (2, \pi/4, 2)$

(d) $\mathbf{D} = \hat{\mathbf{x}}y^2/(x^2+y^2) - \hat{\mathbf{y}}x^2/(x^2+y^2) + \hat{\mathbf{z}}4$ at $P_4 = (1,-1,2)$

第 3-4 节到 3-7 节：梯度、散度和卷曲运算符

3.36 求下列标量函数的梯度：

(a) $T = 3/(x^2+z^2)$b) $V = xy^2z^4$c) $U = z\cos\phi/(1+r^2)$d) $W = e^{-R}$

(((*(

(一个) $10 + x$，对于 $-10 \le x \le 10$

*(b) $T = x^2$，对于 $-10 \le x \le 10$

(b) $T = 100 + xy$，对于 $-10 \le x \le 10$

(c) $T = x^2y^2$，对于 $-10 \le x, y \le 10$

(e) $T = 20 + x + y$，对于 $-10 \le x, y \le 10$

(f) $T = 1 + \sin(\pi x/3)$，对于 $-10 \le x \le 10$

*(g) $T = 1 + \cos(\pi x/3)$, for $-10 \le x \le 10$

(h) $T = 15 + r\cos\phi$，对于 $\begin{cases} 0 \le r \le 1 \\ 0\ 0 \le \phi \le 2\pi \end{cases}$。(i) $T = 15 + r\cos^2\phi$，对于 $\begin{cases} 0 \le r \le 10\ 0 \le \phi \le 2\pi \end{cases}$。*

3.38 标量函数 T 的梯度由下式给出

$$\nabla T = \hat{\mathbf{z}}e^{-3z}$$

如果 $T = 10$ 在 $z = 0$ 处，求 T (z)。

3.39 遵循类似于公式 3 的程序。（3.82）推导出式（3.82）给出的表达式。(3.83) 对于球坐标中的 ∇。

*3.40 对于标量函数 $V = xy^2 - z^2$，确定其沿向量 $A = (\hat{x} - \hat{y}z)$ 方向的方向导数，然后在 $P = (1, -1, 4)$ 处对其求值。

3.41 计算 $E = \hat{x}\,x - \hat{y}\,y$ 沿图 P3.41 所示圆形路径的线段 P_1 至 P_2 的线积分。

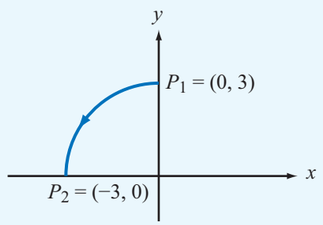


Figure P3.41 Problem 3.41.

3.42 For the scalar function $T = \frac{1}{2}\, e^{-r/5} \cos\phi$, determine its directional derivative along the radial direction $\hat{\mathbf{r}}$ and then evaluate it at $P = (2, \pi/4, 3)$.

*__3.43__ For the scalar function $U = \frac{1}{R}\sin^2\theta$, determine its directional derivative along the range direction $\hat{\mathbf{R}}$ and then evaluate it at $P = (5, \pi/4, \pi/2)$.

3.44 Each of the following vector fields is displayed in **Fig. P3.44** in the form of a vector representation. Determine $\nabla \cdot \mathbf{A}$ analytically and then compare the result with your expectations on the basis of the displayed pattern.

(a) $\mathbf{A} = -\hat{\mathbf{x}}\cos x \sin y + \hat{\mathbf{y}}\sin x \cos y$, for $-\pi \le x, y \le \pi$

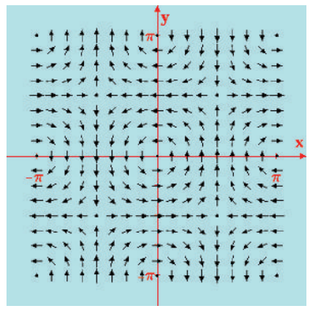


Figure P3.44(a)

(b) $\mathbf{A} = -\hat{\mathbf{x}}\sin 2y + \hat{\mathbf{y}}\cos 2x$, for $-\pi \le x, y \le \pi$

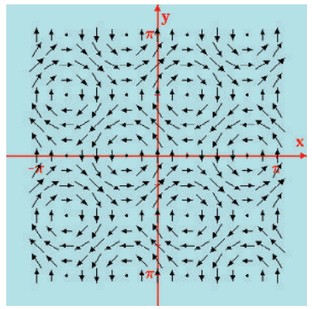


Figure P3.44(b)

(c) $\mathbf{A} = -\hat{\mathbf{x}}xy + \hat{\mathbf{y}}y^2$, for $-10 \le x, y \le 10$

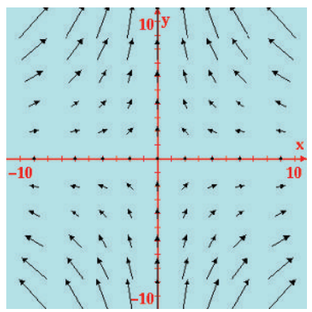


Figure P3.44(c)

(d) $\mathbf{A} = -\hat{\mathbf{x}}\cos x + \hat{\mathbf{y}}\sin y$, for $-\pi \le x, y \le \pi$

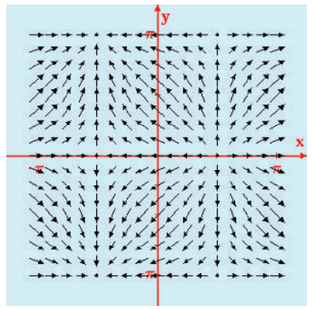


Figure P3.44(d)

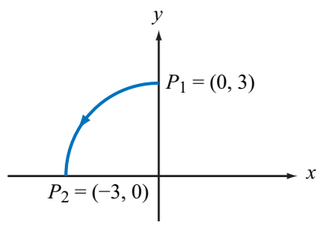


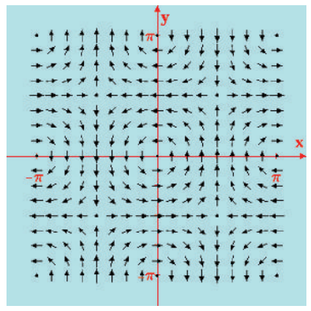
图 P3.41 问题 3.41。

3.42 对于标量函数 $T = \frac{1}{2}\, e^{-r/5} \cos\phi$，确定其沿径向方向 $\hat{r}$ 的方向导数，然后在 $P = (2、\pi/4、3)$ 处对其求值。

*3.43 对于标量函数 $U = \frac{1}{R}\sin^2\theta$，确定其沿范围方向 $\hat{R}$ 的方向导数，然后在 $P = (5、\pi/4、\pi/2)$ 处对其求值。

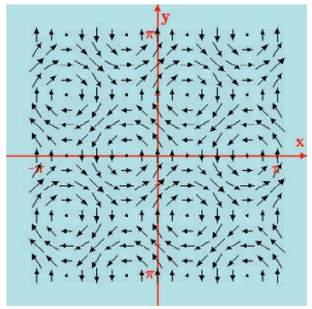
3.44 以下每个向量场都以向量表示的形式显示在图 P3.44 中。通过分析确定 $\nabla \cdot A$，然后根据显示的模式将结果与您的期望进行比较。

(a) $A = -\hat{x}\cos x \sin y + \hat{y}\sin x \cos y$，对于 $-\pi \le x, y \le \pi$



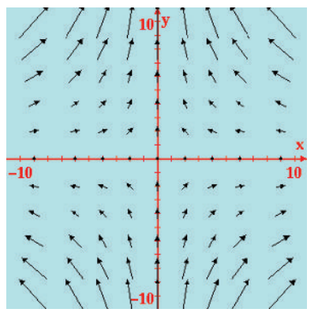
图P3.44(a)

(b) $A = -\hat{x}\sin 2y + \hat{y}\cos 2x$，对于 $-\pi \le x, y \le \pi$



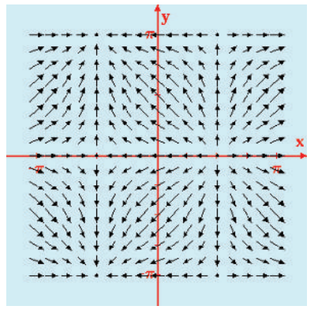
图P3.44(b)

(c) $A = -\hat{x}\, xy + \hat{y}y^2$，对于 $-10 \le x, y \le 10$



图P3.44(c)

(d) $A = -\hat{x}\cos x + \hat{y}\sin y$，对于 $-\pi \le x, y \le \pi$



图P3.44(d)

(e) $\mathbf{A} = \hat{\mathbf{x}}x$, for $-10 \le x \le 10$

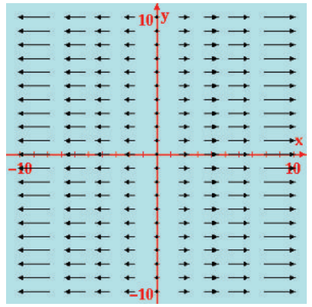


Figure P3.44(e)

(f) $\mathbf{A} = \hat{\mathbf{x}}xy^2$, for $-10 \le x,y \le 10$

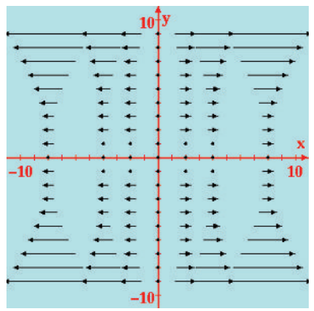


Figure P3.44(f)

(g) $\mathbf{A} = \hat{\mathbf{x}}xy^2 + \hat{\mathbf{y}}x^2y$, for $-10 \le x,y \le 10$

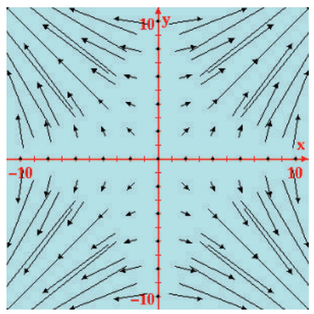


Figure P3.44(g)

(h) $\mathbf{A} = \hat{\mathbf{x}}\sin\left(\frac{\pi x}{10}\right) + \hat{\mathbf{y}}\sin\left(\frac{\pi y}{10}\right)$, for $-10 \le x,y \le 10$

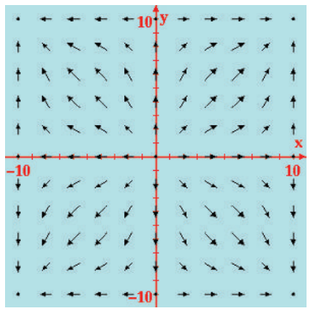


Figure P3.44(h)

(i) $\mathbf{A} = \hat{\mathbf{r}}r + \hat{\boldsymbol{\phi}}r\cos\phi$, for $\begin{cases} 0 \le r \le 10 \\ 0 \le \phi \le 2\pi. \end{cases}$

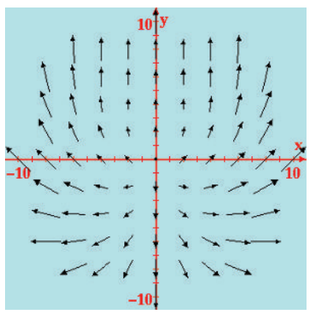


Figure P3.44(i)

(j) $\mathbf{A} = \hat{\mathbf{r}}r^2 + \hat{\boldsymbol{\phi}}r^2\sin\phi$, for $\begin{cases} 0 \le r \le 10 \\ 0 \le \phi \le 2\pi. \end{cases}$

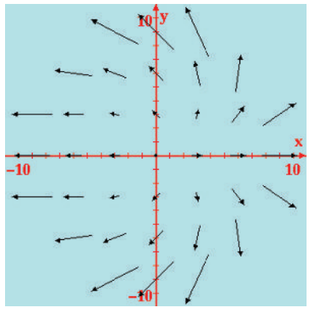
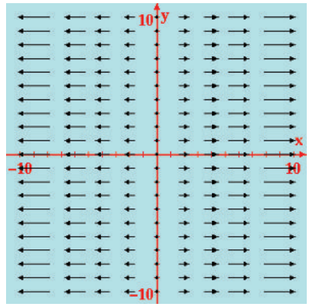


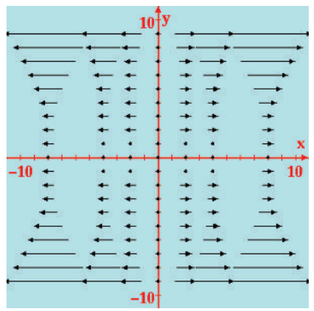
Figure P3.44(j)

(e) $\mathrm{A} = \hat{\mathrm{x}}\,\mathrm{x}$，对于 $-10 \le \mathrm{x} \le 10$



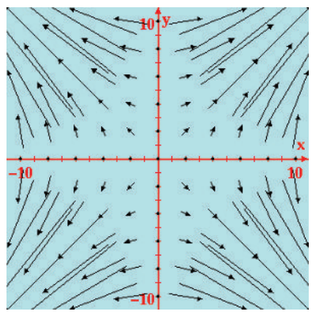
图P3.44(e)

(f) $\mathrm{A} = \hat{\mathrm{x}}\,\mathrm{xy}^2$，对于 $-10 \le \mathrm{x}, \mathrm{y} \le 10$



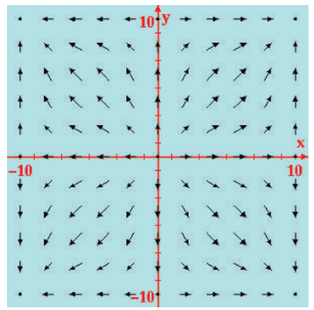
图P3.44(f)

(g) $\mathrm{A} = \hat{\mathrm{x}}\,\mathrm{xy}^2 + \hat{\mathrm{y}}\,\mathrm{x}^2\mathrm{y}$，对于 $-10 \le \mathrm{x}$，$\mathrm{y} \le 10$



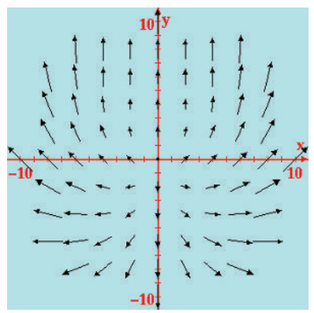
图P3.44(g)

(h) $\mathrm{A} = \hat{\mathrm{x}}\,\sin\left(\frac{\pi x}{10}\right) + \hat{\mathrm{y}}\,\sin\left(\frac{\pi y}{10}\right)$，对于 $-10 \le \mathrm{x}, \mathrm{y} \le 10$



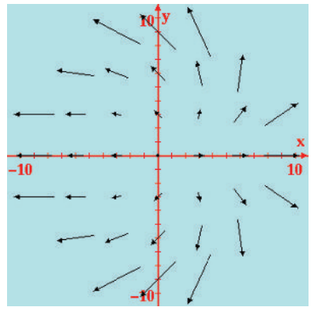
图P3.44(h)

(i) $\mathrm{A} = \hat{\mathrm{r}}\,\mathrm{r} + \hat{\boldsymbol{\phi}}\,\mathrm{r}\cos\phi$，对于 $\begin{cases} 0 \le \mathrm{r} \le 10 \\ 0 \le \phi \le 2\pi \end{cases}$。



图P3.44(i)

(j) $\mathrm{A} = \hat{\mathrm{r}}\,\mathrm{r}^2 + \hat{\boldsymbol{\phi}}\mathrm{r}^2\sin\phi$，对于 $\begin{cases} 0 \le \mathrm{r} \le 10 \\ 0 \le \phi \le 2\pi \end{cases}$。



图P3.44(j)

3.45 The scalar function V is given by

$$V = 2\sin x + xy^2 + ye^{2z}.$$

Determine ∇V and then evaluate it at $P(0,1,2)$.

3.46 The scalar function V is given by

$$V = \frac{2z}{x^2 + y^2}.$$

(a) Determine ∇V in Cartesian coordinates.

(b) Convert the result of part (a) from Cartesian to cylindrical coordinates.

(c) Convert the expression for V into cylindrical coordinates and then determine ∇V in those coordinates. Compare the results of parts (b) and (c).

*3.47 Vector field $\mathbf{E}$ is characterized by the following properties: (a) $\mathbf{E}$ points along $\hat{\mathbf{R}}$; (b) the magnitude of $\mathbf{E}$ is a function of only the distance from the origin; (c) $\mathbf{E}$ vanishes at the origin; and (d) $\nabla \cdot \mathbf{E} = 12$, everywhere. Find an expression for $\mathbf{E}$ that satisfies these properties.

3.48 For the vector field $\mathbf{E} = \hat{\mathbf{x}}xz - \hat{\mathbf{y}}yz^2 - \hat{\mathbf{z}}xy$, verify the divergence theorem by computing

(a) the total outward flux flowing through the surface of a cube centered at the origin and with sides equal to 2 units each and parallel to the Cartesian axes, and

(b) the integral of $\nabla \cdot \mathbf{E}$ over the cube's volume.

3.49 For the vector field $\mathbf{E} = \hat{\mathbf{r}}10e^{-r} - \hat{\mathbf{z}}3z$, verify the divergence theorem for the cylindrical region enclosed by $r = 2$, $z = 0$, and $z = 4$.

*3.50 A vector field $\mathbf{D} = \hat{\mathbf{r}}r^3$ exists in the region between two concentric cylindrical surfaces defined by $r = 1$ and $r = 2$ with both cylinders extending between $z = 0$ and $z = 5$. Verify the divergence theorem by evaluating the following:

(a) $\oint_S \mathbf{D} \cdot d\mathbf{s}$

(b) $\int_v \nabla \cdot \mathbf{D}\, dv$

3.51 For the vector field $\mathbf{D} = \hat{\mathbf{R}}3R^2$, evaluate both sides of the divergence theorem for the region enclosed between the spherical shells defined by $R = 1$ and $R = 2$.

3.52 For the vector field $\mathbf{E} = \hat{\mathbf{x}}xy - \hat{\mathbf{y}}(x^2 + 2y^2)$, calculate

(a) $\oint_C \mathbf{E} \cdot d\mathbf{l}$ around the triangular contour shown in Fig. P3.52(a).

(b) $\int_S (\nabla \times \mathbf{E}) \cdot d\mathbf{s}$ over the area of the triangle.

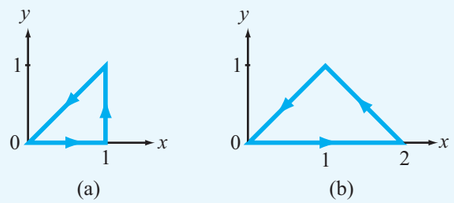


Figure P3.52 Contours for (a) Problem 3.52 and (b) Problem 3.53.

3.53 Repeat Problem 3.52 for the contour shown in Fig. P3.52(b).

3.54 Verify Stokes's theorem for the vector field

$$\mathbf{B} = (\hat{\mathbf{r}}r\cos\phi + \hat{\boldsymbol{\phi}}\sin\phi)$$

by evaluating

(a) $\oint_C \mathbf{B} \cdot d\mathbf{l}$ over the semicircular contour shown in Fig. P3.54(a), and

(b) $\int_S (\nabla \times \mathbf{B}) \cdot d\mathbf{s}$ over the surface of the semicircle.

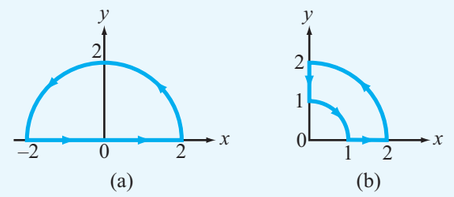


Figure P3.54 Contour paths for (a) Problem 3.54 and (b) Problem 3.55.

3.55 Repeat Problem 3.54 for the contour shown in Fig. P3.54(b).

3.56 Verify Stokes's theorem for the vector field

$$\mathbf{A} = \hat{\mathbf{R}}\cos\theta + \hat{\boldsymbol{\theta}}\sin\theta$$

by evaluating it on the hemisphere of unit radius.

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3.45 标量函数 V 由下式给出

$$V = 2\sin x + xy^2 + ye^{2z}.$$

确定 ∇V，然后在 P(0, 1, 2) 处对其进行评估。

3.46 标量函数 V 由下式给出

$$V = \frac{2z}{x^2 + y^2}.$$

(a) 确定笛卡尔坐标中的 ∇V。
(b) 将(a)部分的结果从笛卡尔坐标转换为柱坐标。 (c) 将 V 的表达式转换为柱坐标，然后在该坐标中确定 ∇V。比较 (b) 和 (c) 部分的结果。

*3.47 矢量场 E 具有以下性质： (a) E 沿 R̂ 指向； (b) E 的大小仅是距原点距离的函数； (c) E 在原点消失； (d) ∇· E = 12，无处不在。找到满足这些属性的 E 表达式。

3.48 对于向量场E = x̂xz − ŷyz² − ẑxy，通过计算验证散度定理
(a) 流过以原点为中心、每条边等于 2 个单位且平行于笛卡尔轴的立方体表面的总向外通量，以及 (b) ∇· E 对立方体体积的积分。

3.49 对于向量场 E = r̂10e^{−r} − ẑ3z，验证 r = 2、z = 0 和 z = 4 围成的圆柱区域的散度定理。

*3.50 矢量场 D = r̂r³ 存在于由 r = 1 和 r = 2 定义的两个同心圆柱面之间的区域中，两个圆柱面都在 z = 0 和 z = 之间延伸 5. 通过评估以下内容来验证散度定理：

(a) $\oint_S \mathbf{D} \cdot d\mathbf{s}$

(b) $\int_v \nabla \cdot \mathbf{D}\, dv$

3.51 对于向量场 D = R̂3R²，计算由 R = 1 和 R = 2 定义的球壳之间包围的区域的散度定理两边。

3.52 对于矢量场 E = x̂xy − ŷ(x² + 2y²)，计算图 P3.52(a) 所示三角形轮廓周围的 (a) $\oint_C$ E · dl 。 (b) 三角形面积上的 $\int_S (\nabla \times E) \cdot ds$。

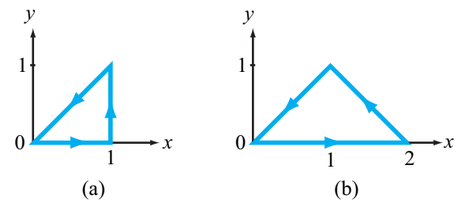


图 P3.52 (a) 问题 3.52 和 (b) 问题 3.53 的等值线。

3.53 对图 P3.52(b) 所示的轮廓重复问题 3.52。

3.54 验证向量场的斯托克斯定理

$$\mathbf{B} = (\hat{\mathbf{r}}r\cos\phi + \hat{\boldsymbol{\phi}}\sin\phi)$$

通过评估
(a)图P3.54(a)所示半圆轮廓上的 $\int_C$ B·dl，以及(b)半圆表面上的 $\int_S (\nabla \times B) \cdot ds$。

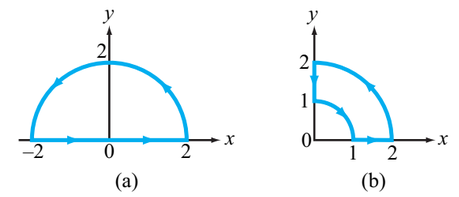


图 P3.54 (a) 问题 3.54 和 (b) 问题 3.55 的轮廓路径。

3.55 对图 P3.54(b) 所示的轮廓重复问题 3.54。

3.56 验证向量场的斯托克斯定理

$$\mathbf{A} = \hat{\mathbf{R}}\cos\theta + \hat{\boldsymbol{\theta}}\sin\theta$$

通过在单位半径的半球上对其进行评估。